

Transtiff: A Stick Interface with Various Stiffness by Artificial Muscle Mechanism

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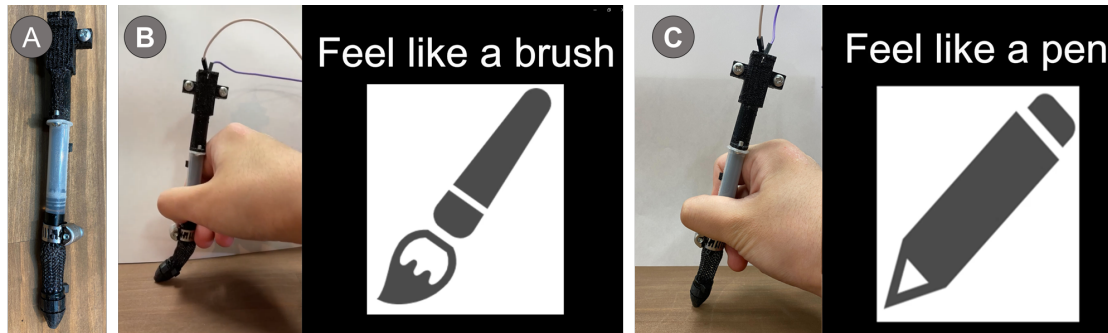


Figure 1: Transtiff: (A) Prototype (B) Soft mode feels like a brush. (C) Hard mode feels like a pen.

ABSTRACT

We manipulate stick objects, such as chopsticks and pens in our daily life. The senses of the human hand are extremely sensitive and can acquire detailed information. By perceiving changes in the feel of the finger, we perceive the characteristic of the object when we manipulate sticks, such as pens and brushes. Therefore, we can extend the tactile experience of touching something by controlling a stick's grasping sensation. In this study, we propose *Transtiff* which has a joint that changes its stiffness dynamically to a stick object that generally cannot bend. We use a piston mechanism that uses a small motor to compress the liquid in a flexible tube to control the stiffness of the joint.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction devices; Haptic devices.

KEYWORDS

HCI, haptic device, bending, pen interaction, fluid-based interface

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1 INTRODUCTION

In recent years, there has been much research on tactile presentation with the advancement of VR technology. In particular, many devices have been proposed to create the sensation of directly touching objects in virtual space [1–5]. These devices presented the sensations vary widely in size, shape, weight, touch, and stiffness of the object and presented different elements for purposes. Since many brain regions are allocated to human hand manipulation [6], we can distinguish and perceive various tactile sensations. Humans can obtain detailed information from the sensation of grasping when they touch an object with sticks. Thus we can extend the tactile experience through techniques that control the sensation of grasping. In this study, we propose *Transtiff* (Figure 1), which has a stiffness-changing joint that can be applied to the relay part of a stick device, aiming to control the tactile sensation of the object touched through the stick by changing the stiffness of the device itself. Generally, stiffness-changing joints are used as robotic joints [7]. However, they are difficult to apply to everyday devices such as pens because of the large size and weight of the device. Therefore, We realize *Transtiff* with a stiffness-changing tip by using the mechanism of the McKibben artificial muscle [8], in which a liquid is pressurized in a silicone rubber tube, and the expansion is received by a braided tube.

2 TRANSTIFF

Figure 2 shows the structure of *Transtiff*. Our goal is to control the stiffness of the silicon tube that forms a part of the device, thereby changing the sensation of stiffness felt by the user grasping *Transtiff*. The difference compared to existing tactile devices is that *Transtiff* changes its behavior by controlling the stiffness of the joints rather than directly presenting tactile sensations to the user's hand or fingers. As a result, *Transtiff* changes the tactile sensation felt by the user. *Hydraio*[9] is a pen-type device that uses liquid. This device controls the stiffness of the water bag that touches the

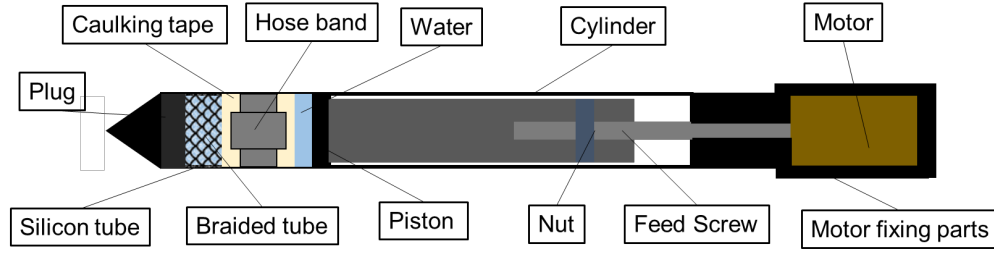


Figure 2: Structure of Transtiff.

index finger, so the tactile sensation is limited to the index finger. On the other hand, our method can change the tactile sensation of the object through Transtiff, regardless of how to grip. Although we use a wide variety of stick objects in our daily lives, we focused on a pen-shaped device because it is used frequently and can be used without discomfort, even if its diameter is somewhat large. We aim to achieve a change in stiffness that the user can perceive without compromising the pen-like characteristics of being light and easy to hold.

3 IMPLEMENTATION A PROTOTYPE

Figure 3 shows the mechanism of the stiffness increase. Rotation of a feed screw connected to a motor (DC motor, 60 rpm) causes a nut to move linearly, enabling the reciprocating motion of an attached piston. This pushes the liquid in the cylinder toward the silicone tube. The tube section is equipped with a McKibben-type artificial muscle mechanism. The piston compresses water into the silicone tube, and the braided tube holds the tube from expanding. This causes the tube to expand in the radial direction, increasing the overall stiffness. To prevent liquid leakage, we inserted a plug-in shape of a pen nib on the tip side, wrapped caulking tape around the cylinder side, and tightened it with a hose band. We self-made the plug, piston, and motor fixing parts using a 3D printer. We achieved the piston action by switching the direction of rotation of the motor. In the current prototype, we control the direction of rotation by operating scroll bars displayed on the GUI on the PC screen. Figure 4 shows an overall image of the prototype. Currently, we are investigating a more precise cylinder and a method of strictly sealing the liquid. Once the prototype is sufficiently robust, we plan to conduct evaluation experiments. We are planning to investigate the quantitative change in piston push length and stiffness, as well as subjective sensory change in stiffness.

4 CONCLUSION AND FUTURE WORK

We proposed Transtiff, a stick device with joint stiffness control. Using this system, we intend to create applications that enable users to experience the perception of multiple stiffness sensations with a single device. For example, assuming that Transtiff is used for writing, the user can use it as a pen when it is stiff and as a brush when it is flexible; FlexStylus [10] has shown that flexible stylus pens have a positive impact on input tasks, and from this, we can find the usefulness of our proposed feature of softening stick devices. Furthermore, Piovarči et al. [11] propose a data-driven simulation method that can aid in designing a new stylus, and

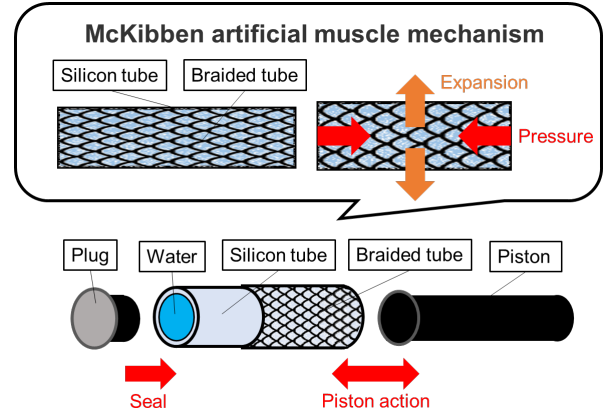


Figure 3: Mechanism of the stiffness increase.

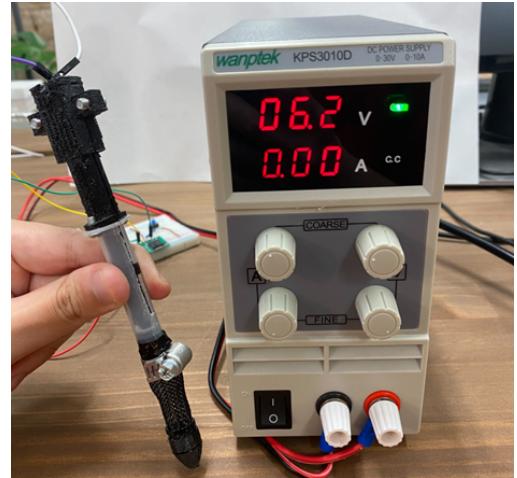


Figure 4: Overall of prototype.

from this, we can use it to help design pen devices. Future work includes (i) consideration of pen tips (actual pen or stylus pen); (ii) automation of motor drive; and so on.

ACKNOWLEDGMENTS

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